

Growth Kinetics of Cell Cultures

Chapter 7 — from unstructured black-box kinetics to structured, segregated and mechanistic models

Contents

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- 7.1 Model structure and complexity
 - 7.2 A general structure for kinetic models
 - 7.3 Unstructured growth kinetics
 - 7.4 Simple structured models
 - 7.5 Fraction of repressor-free operators
 - 7.6 Morphologically structured models
 - 7.7 Transport through the cell membrane
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7.1

1 Model structure and complexity complexity

The first step is an oral model

“In a series of steady-state chemostat experiments, the specific rate of biomass production r_x is linear in the glucose concentration s for small s , but for large s , r_x is independent of s . This indicates that the rate of the bioreaction is first order in s for $s \rightarrow 0$ like any other chemical reaction and that the rate eventually becomes constant because all the metabolic machinery in the cell is fully engaged in converting S .”

These sentences perfectly express the content of the Monod model (7.16). The functionality of $r_x(s)$ is determined afterwards by data fitting — the shape is based on physical arguments.

The aim decides the structure

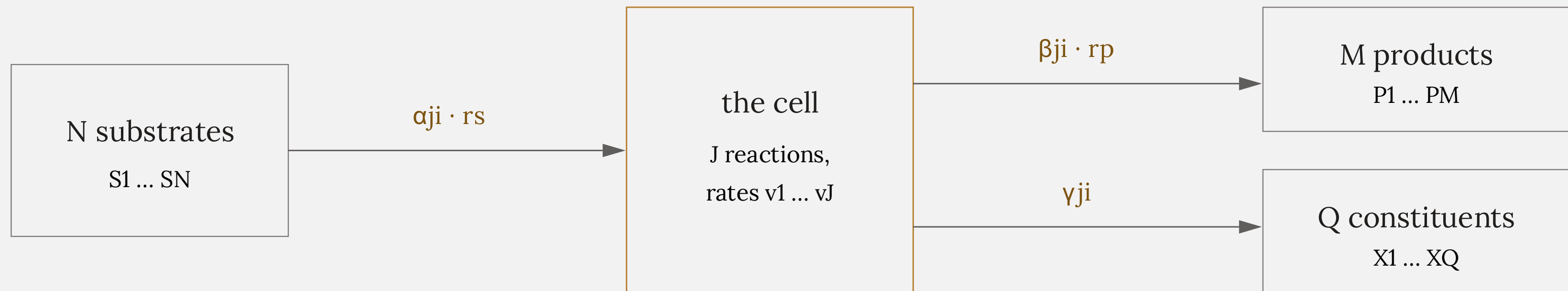
IF THE AIM IS TO...	USE...	SECTION
Simulate biomass in a process	Unstructured, possibly segregated	7.3
Simulate dynamic growth	Simple structured (compartment) models	7.4
Understand the fundamentals	Mechanistic models of one part of metabolism	7.5, 7.7
Investigate population distributions	Morphologically structured, segregated	7.6, Ch. 8

“Keep the model as simple as possible, while including the phenomena that are essential to explain the observed phenomena.”

7.2

A general structure for kinetic models
models

Fig. 7.1 · The general reaction scheme



Stoichiometry of the j th reaction: $\sum \alpha_{ji} S_i + \sum \beta_{ji} P_i + \sum \gamma_{ji} X_i = 0$ (7.1), or in matrix form $AS + BP + \Gamma X = 0$ (7.2). The coefficients are identical to the yield coefficients when the reactions are lumped into one.

Reaction rates and dynamic mass balances

Rates

$$(r_s)_{\text{cons}} = A^T v \quad r_p = B^T v \quad r_x = \Gamma^T v$$

The exchange rates that a fermentation can actually measure are linear in the vector v of intracellular reaction rates (7.7–7.9). The specific growth rate is the sum over all Q constituents, $\mu = r_x = \sum r_{x,i}$ (7.6).

Equation (7.14) is independent of how the reactor is operated.

Balances

$$d(cV)/dt = r(c)xV + v_f c_f - v_e c_e$$

The general balance (7.10) holds for all substrates, products and biomass components. Inside the cell, $dX_i/dt = \Gamma_i^T v - \mu X_i$ (7.14): every constituent must be made at least at the rate μX_i , or growth dilutes it away.

7.3

Unstructured growth kinetics

One rate expression, constant yields

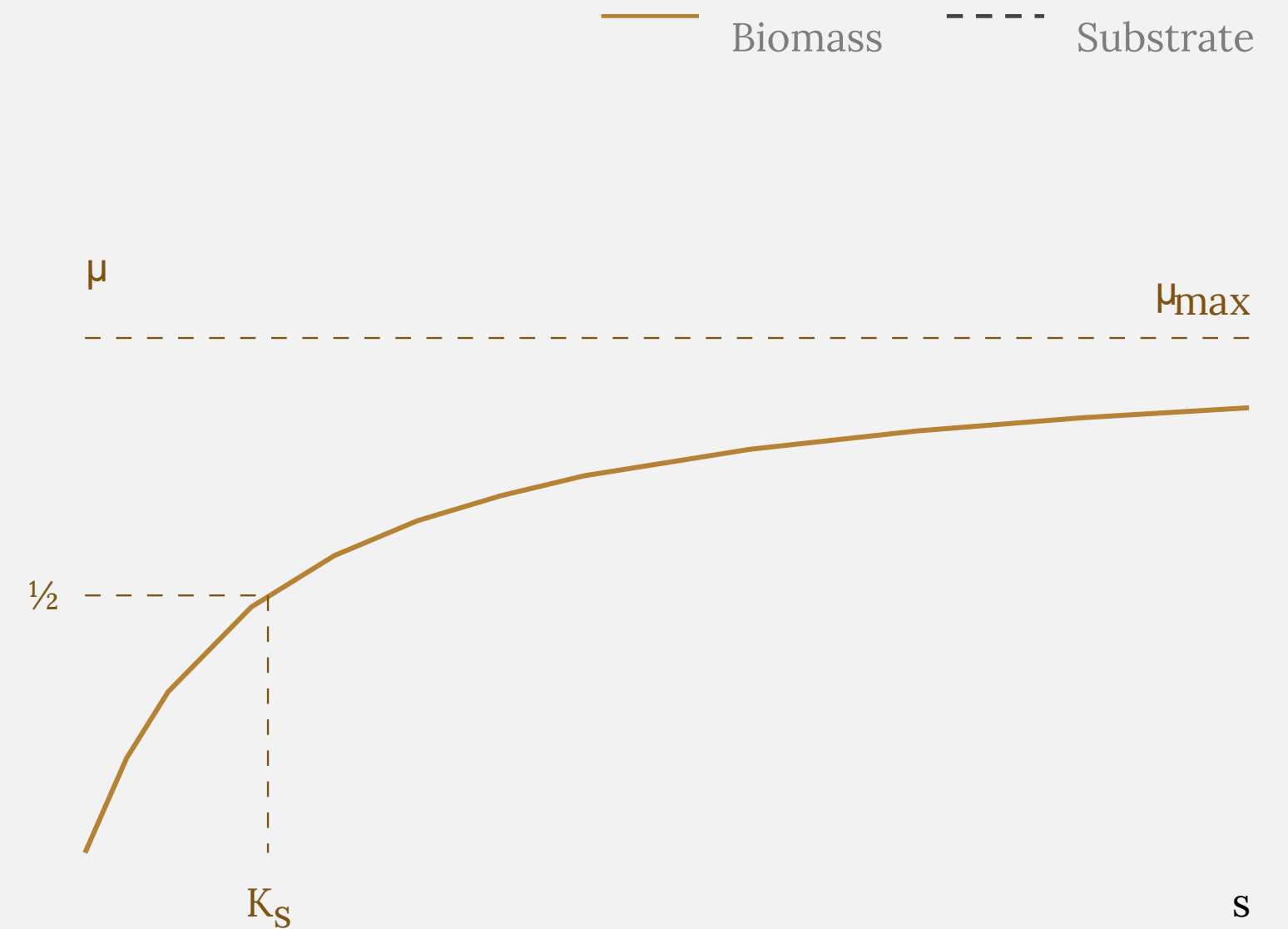
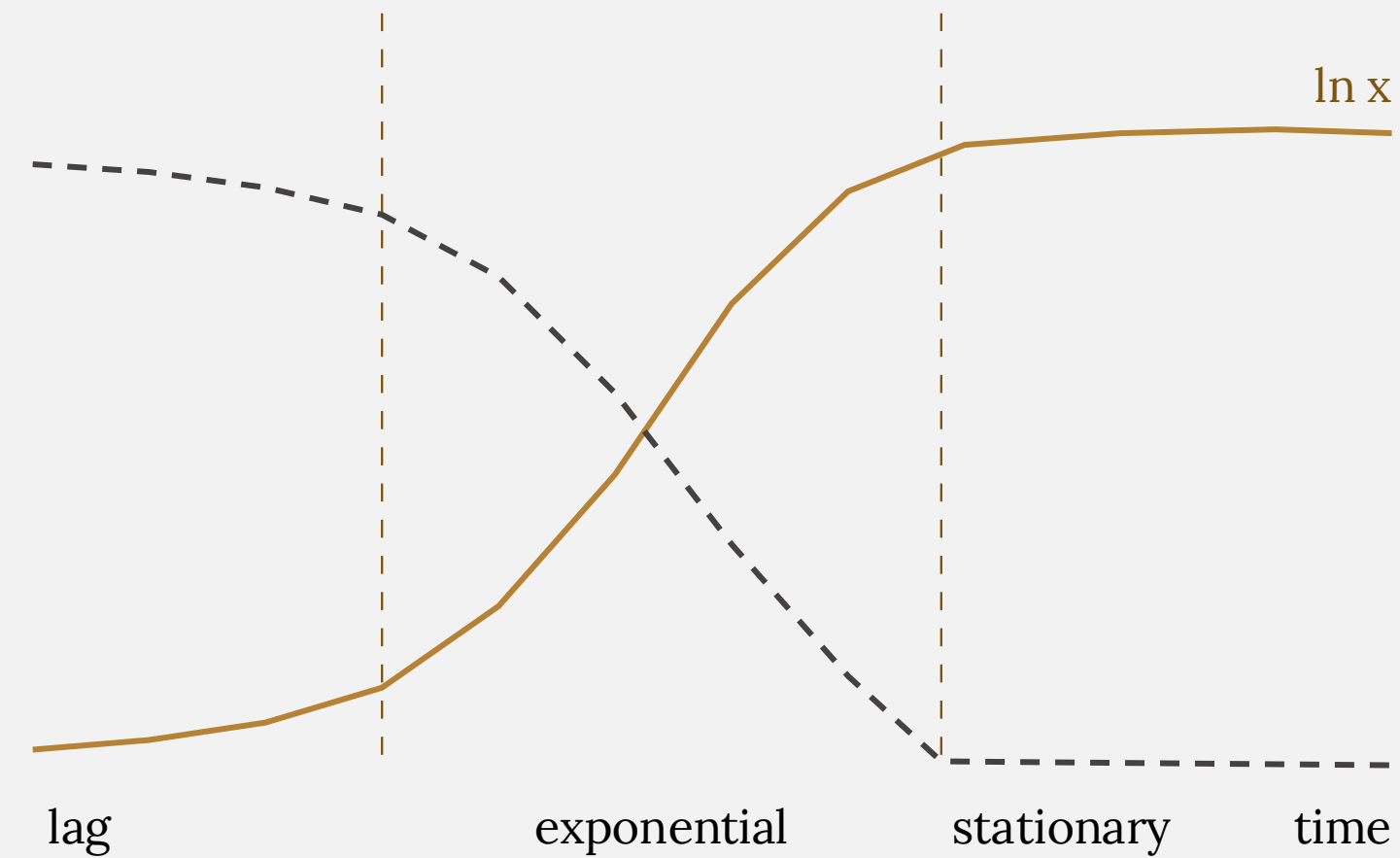
$$-r_s = F(s, p_1, p_2, \text{pH}, T)$$

$$r_x = \mu = Y_{sx}(-r_s) = 0.137(-r_s) \quad r_{p1} = 0.510(-r_s)$$

Equation (7.15) for anaerobic yeast, with the yield coefficients of the black-box model (3.23).

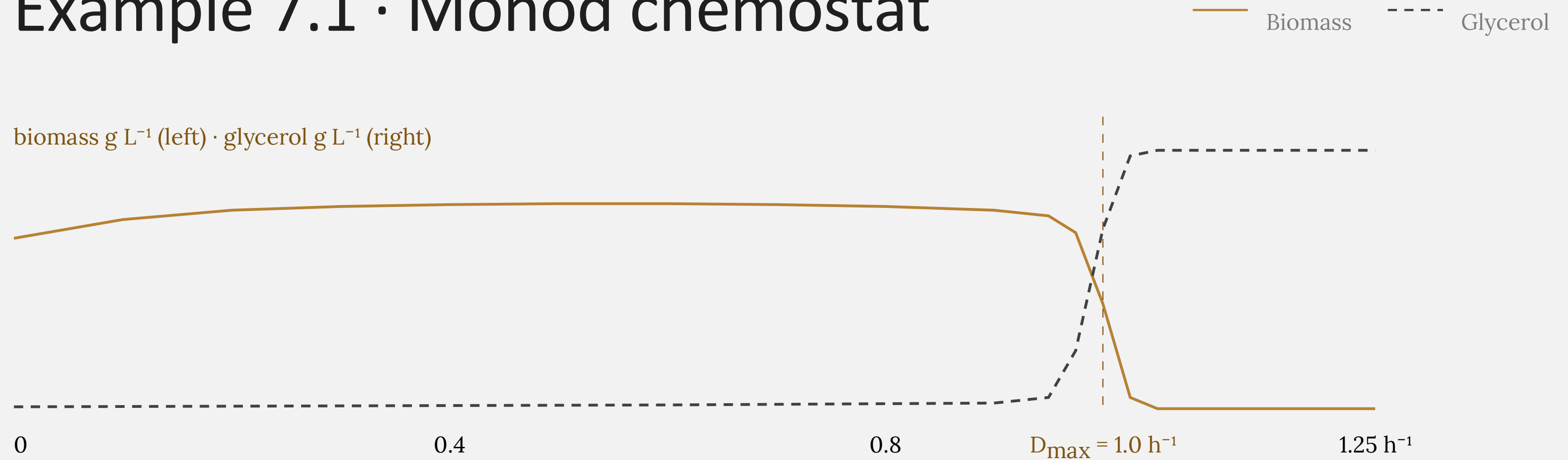
All cellular components are pooled into one biomass concentration x .
Only the key rate is a function of the state variables c , pH and T .
 F extrapolates only as long as the yield coefficients stay constant.

7.3.1 · The Monod model



$r_x = \mu = \mu_{\max} s / (s + K_s)$ (7.16). K_s does not represent the saturation constant for substrate uptake, but only an overall saturation constant for the whole growth process.

Example 7.1 · Monod chemostat



$$r_x = D s = D K_s / (\mu_{\max} - D)$$

$$D_{\max} = \mu_{\max} s_f / (s_f + K_s)$$

Aerobacter aerogenes on glycerol (Herbert 1959): $\mu_{\max} = 1.0 \text{ h}^{-1}$, $K_s = 0.01 \text{ g L}^{-1}$, $Y_{sx} = 0.53 \text{ g DW g}^{-1}$. At low D the model predicts too much biomass — maintenance is missing.

Table 7.1 · K_s values from the literature

Species	Substrate	K_s (mg L ⁻¹)
Aerobacter aerogenes	Glucose	8
Escherichia coli	Glucose	4
Klebsiella aerogenes	Glucose	9
Klebsiella aerogenes	Glycerol	9
Klebsiella oxytoca	Glucose	10
Klebsiella oxytoca	Arabinose	50
Lactococcus cremoris	Glucose	2
Lactococcus cremoris	Lactose	10
Saccharomyces cerevisiae	Glucose	150
Penicillium chrysogenum	Glucose	4
Aspergillus oryzae	Glucose	5

Table 7.2 · Other unstructured expressions

Name	Kinetic expression	Comment
Tessier	$\mu = \mu_{\max}(1 - e^{-s/K_s})$	Approaches $\mu_{\max}$ too fast
Moser	$\mu = \mu_{\max} s^n / (s^n + K_s)$	The Hill form, mechanism absent
Contois	$\mu = \mu_{\max} s / (s + K_s x)$	Inhibition by biomass — doubtful
Blackman	$\mu = \mu_{\max} s / 2K_s, s \leq 2K_s$	Piecewise linear
Logistic law	$\mu = \mu_{\max}(1 - x/K_x)$	Use “positively discouraged”

Inhibition, and more than one substrate

Inhibition

$$\mu = \mu_{\max} s / (s^2/K_i + s + K_s)$$

$$\mu = \mu_{\max} s / (s + K_s) \cdot (1 - p/p_{\max})$$

Substrate inhibition (7.18) describes high oxygen concentrations and toxic compounds in bioremediation. For ethanol, (7.20) is preferred: inhibition sets in abruptly, and wine yeast stops at about 120 g L⁻¹ — improved strains reach above 170.

Several substrates

$$\mu = \mu_{\max,1} \mu_{\max,2} s_1 s_2 / (s_1 + K_1)(s_2 + K_2)$$

$$\mu/\mu_{\max} = \min(s_1/(s_1 + K_1), s_2/(s_2 + K_2))$$

For O₂ and CH₄ in single-cell protein, (7.22) gives only 81% of the maximum rate when each substrate stands at 90% — hardly reasonable, so Roels proposed (7.23) and (7.24). Substitutable substrates need a structured model.

“The genesis of the Monod model:
1942, small batch reactors, six
sugars — $K_s = 4 \text{ mg L}^{-1}$, $t_d = 44 \text{ min}$, μ
 $= 0.94 \text{ h}^{-1}$, $Y_{sx} = 0.233 \text{ g g}^{-1}$.”

— a scientist working at the limits of the equipment of his time

What unstructured kinetics can carry

Constant yields: only the limiting substrate needs a rate expression.
(7.16) and its relatives usually suffice; fit them in a continuous reactor.
Picking the correct limiting substrate is the hard part, not the fitting.
No unstructured model describes fast transients in a stirred tank.

Catabolic reactions respond within seconds; a change in biomass composition has a time constant of one to several hours. Unstructured models underestimate those time constants by orders of magnitude.

7.3.2 · Maintenance as a second reaction

$$-r_s = Y_{xs}^{true} \mu + m_s Y_{sx} = \mu / (Y_{xs}^{true} \mu + m_s)$$

Pirt's linear relation (7.25, 7.27). The yield is no longer constant: it falls to zero as $\mu \rightarrow 0$ and approaches Y_{sx}^{true} at high μ .

Species (glucose, aerobic)	$Y_{xs}^{true} \text{ (g g}^{-1}\text{)}$	$m_s \text{ (g (g DW h)}^{-1}\text{)}$	$m_o \text{ (mmol (g DW h)}^{-1}\text{)}$
Escherichia coli	2.27	0.057	0.45
Klebsiella aerogenes	2.27	0.063	0.99
Saccharomyces cerevisiae	1.85	0.015	0.62
Penicillium chrysogenum	2.17	0.021	0.87
Aspergillus awamori	1.92	0.016	0.62

Example 7.2 · Chemostat with maintenance

$$x = D / (Y_{xs}^{\text{true}} D + m_s) \cdot (s_f - s)$$

s is unchanged from Example 7.1, since $\mu = D$ at steady state in both cases. The biomass concentration is not.

Revised parameters: $\mu_{\text{max}} = 1.0 \text{ h}^{-1}$, $K_s = 0.01 \text{ g L}^{-1}$, $m_s = 0.08$, $Y_{sx} = 0.55$.

Observed yield 0.50 and 0.53 at $D = 0.4$ and 0.8 h^{-1} — as before.

At $D = 0.1 \text{ h}^{-1}$ the yield is 0.37, now in agreement with the data.

Example 7.3 · The respiratory bottleneck

Two reactions, one funnel

$$v_{\text{res}} = \min(r_s, r_{o,\text{max}}/\alpha_{12})$$

$$v_{\text{fer}} = |r_s| - r_{o,\text{max}}/\alpha_{12}$$

Above the critical glucose uptake rate, the excess flux overflows to ethanol — the Crabtree effect. $\mu = \gamma_1 v_{\text{res}} + \gamma_2 v_{\text{fer}}$, and the biomass yield drops from $\gamma_1 = 0.49$ to a mixture with $\gamma_2 = 0.05$.

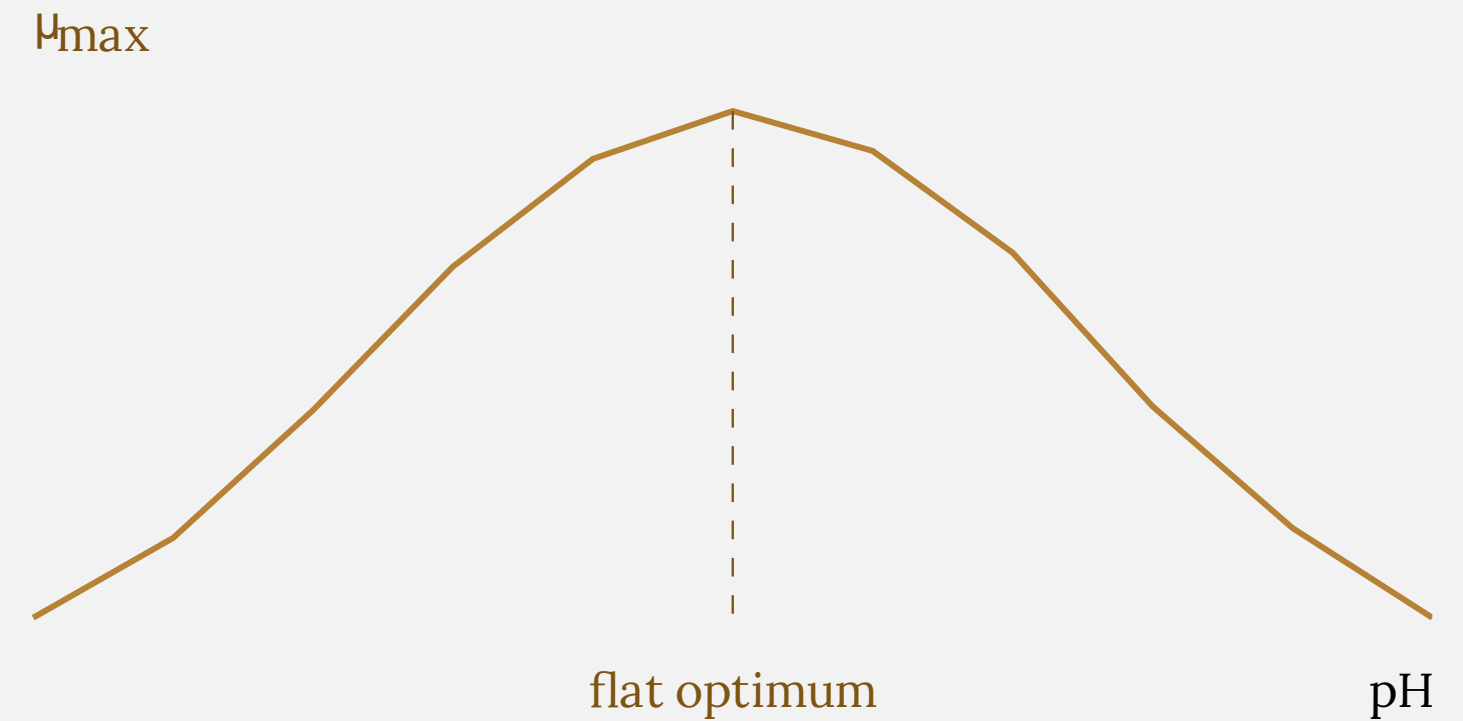
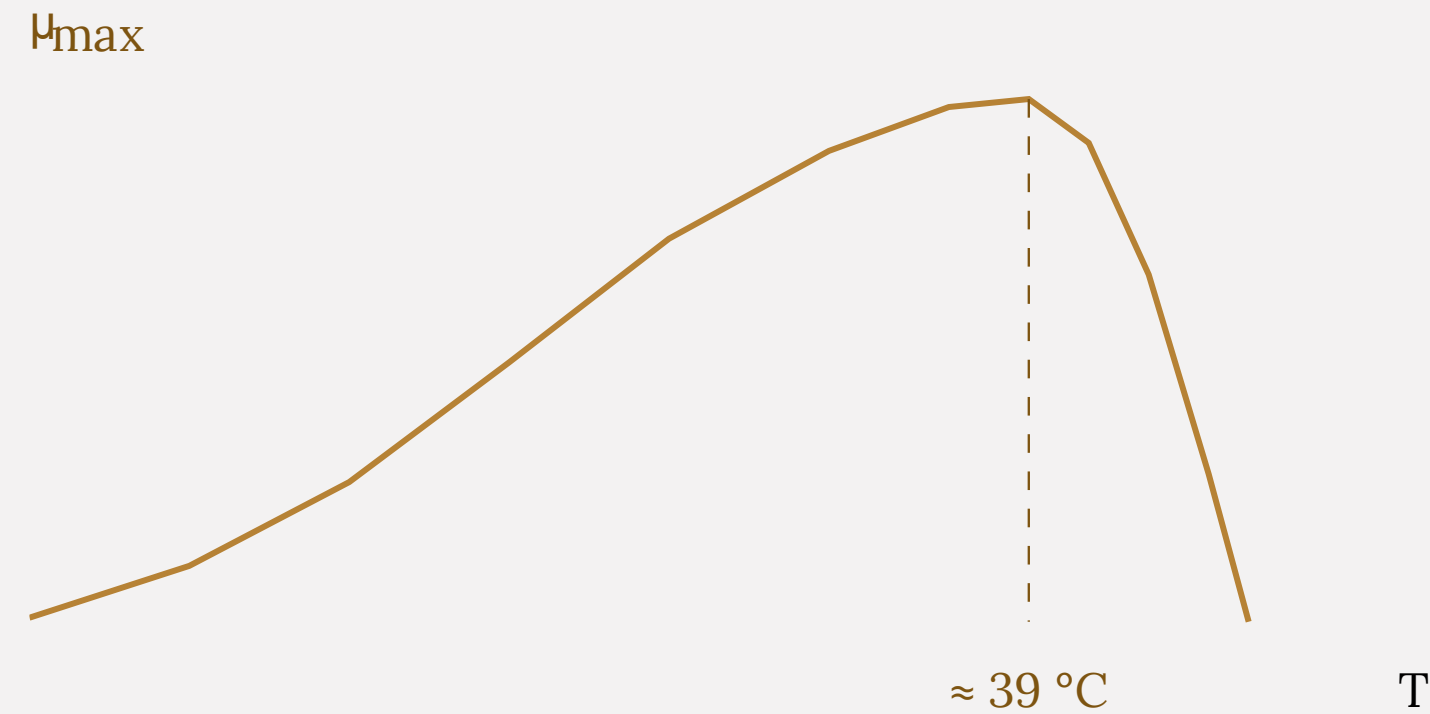
RQ as the control handle

$$\text{RQ} = r_{\text{CO}_2} / r_{\text{O}_2}$$

Without fermentative metabolism $\text{RQ} = \beta_{11}/\alpha_{12} \approx 1$; above D_{crit} it rises above 1. Head-space gas analysis therefore gives an almost continuous signal for controlling the glucose feed in baker's yeast production.

Example 7.4 extends the model to proteinase A: the volumetric productivity has a sharp maximum exactly at D_{crit} , so the process must be run at or very close to it.

7.3.3 · Temperature and pH



$$\mu_{\max} = A e^{-E_g/RT} / (1 + B e^{-\Delta G_d/RT})$$

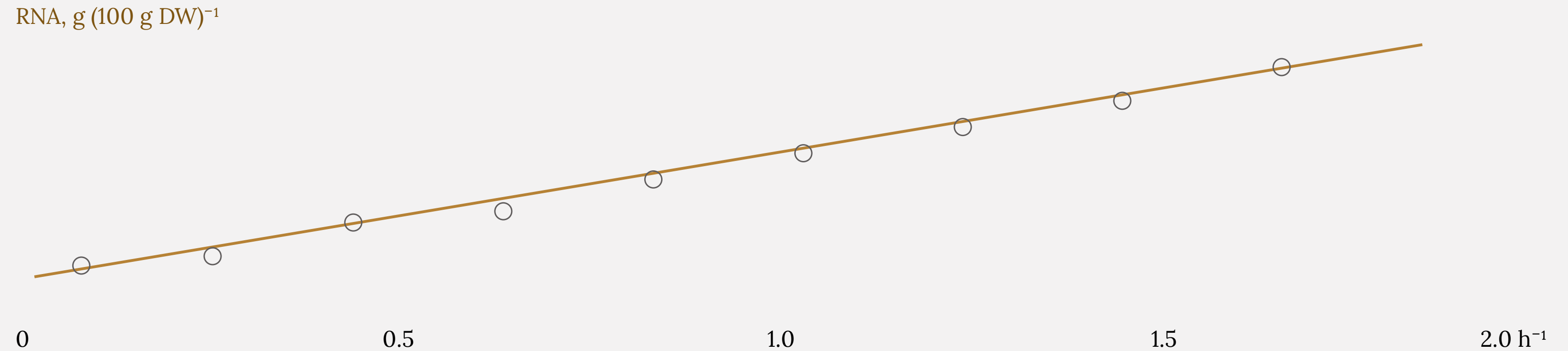
$$\mu_{\max} = k e_{\text{tot}} / (1 + [\text{H}^+]/K_1 + K_2/[\text{H}^+])$$

E. coli: $E_g = 58\text{ kJ mol}^{-1}$ on both rich and minimal medium — A carries the medium, E_g the strain. *A. oryzae*: $K_1 = 4 \cdot 10^{-3}$, $K_2 = 2 \cdot 10^{-8}$, $k e_{\text{tot}} = 0.3\text{ h}^{-1}$.

7.4

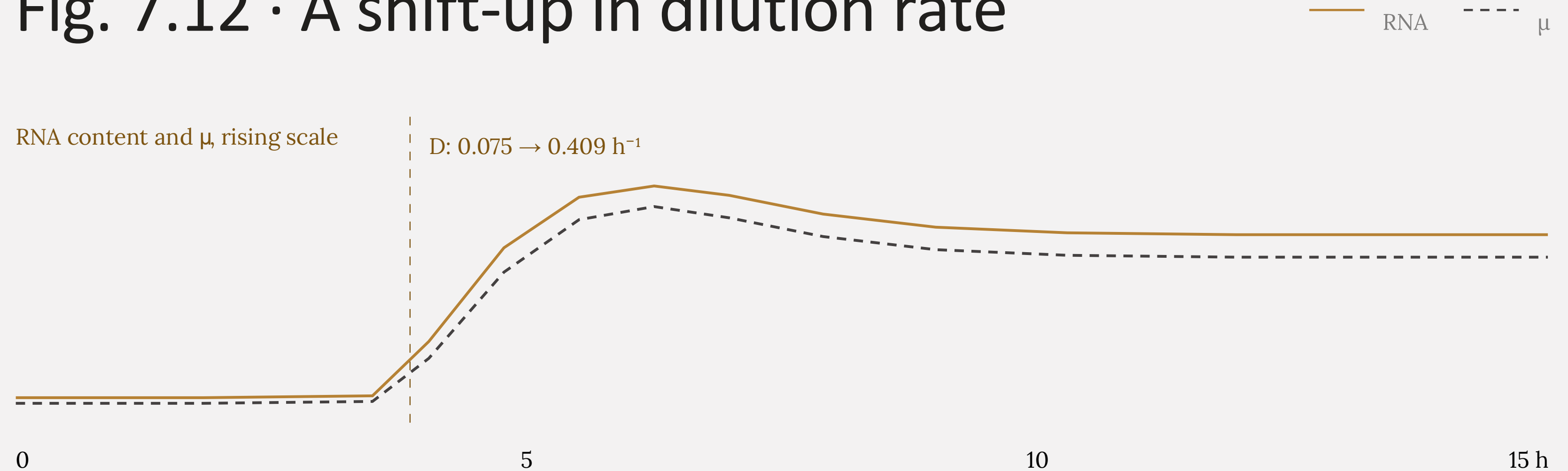
Simple structured models

7.4.1 · RNA measures the active compartment



E. coli, *A. chroococcum*, *A. aerogenes*, *S. typhimurium*, *B. megatarium*, *C. utilis*, *P. chrysogenum*. Ribosomes are 60% rRNA, and rRNA is over 80% of the stable RNA — so an RNA measurement reads the ribosome level directly.

Fig. 7.12 · A shift-up in dilution rate



Lactococcus cremoris, glucose-limited (Benthin et al. 1991). μ mimics the RNA profile with an overshoot before settling. Glucose goes from 5.5 mg L⁻¹ through an excursion to 0.75 g L⁻¹ and back to 10 mg L⁻¹ — the RNA content depends dramatically on s in that range.

Two compartments, two formulations

Williams (7.32)

$$X_A = (D + k_3) / \gamma_{22} k_2$$

Precursors and ribosomes form the A compartment, macromolecules the G compartment. Example 7.5: X_A rises with D as it should, but the fitted $k_3 = 0.5 \text{ h}^{-1}$ is far too high to be reasonable. Without the decay term the model gives $X_A = 0$ at $D = 0$, and resting cells do keep ribosomes on stand-by.

Lactic acid bacteria (Ex. 7.6)

$$\mu = [\gamma_{11} k_1 f_1 - (1 - \gamma_{22}) k_2 f_2] X_A$$

Both rates are proportional to X_A , and $K_{s,1} > K_{s,2}$ makes the active compartment grow with s . Two wash-out experiments to $D = 0.99 \text{ h}^{-1}$ — from 0.10 and from 0.50 h^{-1} — differ sharply, and the model captures both with parameters fitted to steady states alone.

The model takes correct account of the previous history of the cells — which is the whole point of structure.

Catabolism and anabolism decouple

Anaerobic yeast, D stepped $0.053 \rightarrow 0.25 \text{ h}^{-1}$, with $\mu_{\text{max}} = 0.39 \text{ h}^{-1}$.

Before the step 14% of the machinery was active; after it, up to 64%.

The catabolic rate jumps at once — a hidden capacity, 35% of the change.

Anabolism follows slowly: $\tau_{\text{cat}} = 2.5 \text{ h}$ against $\tau_x = 3.2 \text{ h}$.

Fig. 7.15: on a shift down from 0.49 to 0.095 h^{-1} the culture turns homolactic to heterolactic — LDH activity falls threefold while pyruvate-formate lyase rises by about the same factor. No embellishment of a Monod model could produce either response.

7.4.2 · Cybernetic models

Bread dough contains both glucose and maltose, and leavening has two distinct phases. Sequential uptake follows from catabolite repression; the cybernetic hypothesis is that the cell allocates its machinery so as to optimise growth. One key enzyme is the bottleneck for each substrate, and it must be synthesised before growth on that substrate can occur.

$$u_i = r_i^* / \sum r_j^*$$

$$w_i = r_i^* / \max_j(r_j^*)$$

u allocates resources to enzyme synthesis (the matching law); w controls at the enzyme level so growth runs on the best-suited substrate. All parameters can be fitted from single-substrate experiments — the model then predicts the mixtures.

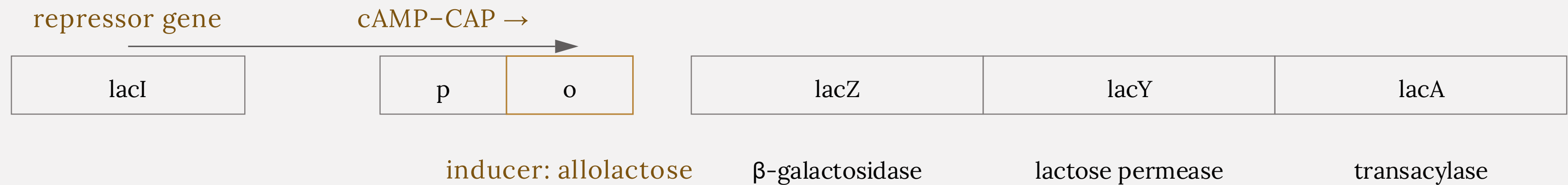
Table 7.6 · Growth on substitutable substrates

Cells prefer the fastest substrate — hence sequential use in batch.
Use turns simultaneous when another substrate becomes limiting.
Even then, μ never exceeds that on any single substrate alone.
Add a faster substrate, and growth on the slower one stops.
In continuous culture: simultaneous at low D, faster substrate at high D.

7.5

Fraction of repressor-free operators

Fig. 7.17 · The lac-operon



Two control mechanisms enter the model: control at the operator by the repressor protein, with lactose as inducer, and carbon catabolite repression at the promoter, with glucose as inhibitor.

Two fractions carry the regulation

Q_1 · repressor-free operators

$$(1 + K_1[S]^n) / (1 + K_1[S]^n + K_2[X_r])$$

From the four binding equilibria (7.39) with $n = 4$ sites, using three simplifications: repressors outnumber operators 10–20 fold, the inducer–repressor complex binds the operator weakly, and inducer is in excess. Read it as the probability that an operator is repressor-free.

Q_2 · activated promoters

$$K_5 K_6 [X_{cAMP}]^m [X_{CAP}] / (1 + \text{same})$$

Carbon catabolite repression: the cAMP–CAP complex destabilises the helix downstream and so promotes polymerase binding. The link to the medium, $X_{cAMP} = K / (K + s_{glc})$ (7.54), is entirely empirical – and it governs the performance of the whole model.

Example 7.7 · A model for diauxic growth

$$\begin{aligned}dX_{\text{mRNA}}/dt &= k_1 Q_1 Q_2 f(\mu) - k_3 X_{\text{mRNA}} - \mu X_{\text{mRNA}} \\dX_E/dt &= k_2 X_{\text{mRNA}} - k_4 X_E - \mu X_E \\ \mu &= \gamma_{\text{glc}} r_{\text{glc}} + \gamma_{\text{lac}} r_{\text{lac}}, \quad r_{\text{lac}} \propto X_E\end{aligned}$$

Both sugars high: Q_1 high, Q_2 low — mRNA and X_E stay negligible.

Glucose falls: cAMP rises, Q_2 rises, enzymes are made de novo.

The lag between the two growth phases follows from the synthesis time.

Protein synthesis as a mechanistic model

$$r_{\text{mRNA}} = k_m \varepsilon_{\text{tr}} X_g = k_m Q_1 Q_2 Q_3 X_g$$

Q_3 is the fraction of promoters complexed with RNA polymerase; X_g is the gene copy number — for recombinant cells, the plasmid copy number.

mRNA elongation $\approx 2,400$ nucleotides min^{-1} per active polymerase.

Polypeptide elongation $\approx 1,200$ amino acids min^{-1} per ribosome.

At $t_d = 1$ h: $k_m = 4.26$ mRNA (gene $\text{min})^{-1}$, $k_p = 13.8$ proteins (mRNA $\text{min})^{-1}$.

mRNA degradation ≈ 0.53 min^{-1} ; protein turnover below 0.1 h^{-1} .

7.6

6 Morphologically structured models

Forms, metamorphosis, and the weighted μ

$$Z_i - Z_j = 0 \quad \Delta Z = 0 \quad \mu = \sum \mu_i Z_i$$
$$dZ_q/dt = \Delta_q^\top u + (\mu_q - \mu) Z_q$$

A finite number Q of cell states, each with its own unstructured kinetics.
Metamorphosis reactions are empirical — mechanisms largely unknown.
They conserve mass: each reaction's coefficients sum to zero.
Three terms: exchange between forms, growth, dilution by growth.

Example 7.8 · Plasmid instability

$$\begin{aligned} dZ_p/dt &= [(1 - \delta)\mu_p - \mu] Z_p \\ dZ_h/dt &= \delta \mu_p Z_p + (\mu_h - \mu) Z_h \end{aligned}$$

δ is the probability that a division yields a plasmid-free daughter cell.

Plasmid-free cells carry a lower metabolic burden, so $\mu_h > \mu_p$.

Both terms in the Z_h balance are positive: formed, and outgrowing.

Even a small tendency ends in a large fraction of non-producing cells.

7.6.1 · Oscillating yeast cultures

Why they can persist

Budding division is asymmetric: a daughter needs t_d to become a mother, then t_m of maturation and t_b of budding. Synchrony survives if the generation time of one subpopulation is an integer multiple of the other — Hjortso and Nielsen get a constant mother-to-daughter ratio of $(1 + \sqrt{5})/2$.

What oscillates

Dissolved oxygen, ethanol and acetate, glucose, the oxygen uptake and carbon dioxide evolution rates, and intracellular NADH and DNA — spontaneously, in glucose-limited chemostats, for many generations. Note 7.7 predicts $T_{osc} = \ln r / D$, inversely proportional to D as observed.

Spontaneous oscillations are never seen in industrial tanks of, say, 80 m³ — which is one reason industry has shown little prolonged interest in them.

7.6.2 · Growth of filamentous microorganisms

Apical, subapical, hyphal

All cells make protoplasm, but extension occurs only at the tips. Branching forms new apical compartments; vacuolated hyphal cells no longer extend. Example 7.9 uses three forms and three metamorphosis reactions — branching, tip growth, differentiation — each first order in the disappearing form.

The hyphal growth unit

$$l_{\text{hgu}} = \mu / a_2 u_1 q_{\text{tip}} = \mu l_{\text{hgu}}$$

Total mycelium length per tip. Rich conditions give a densely branched mycelium with a large hyphal diameter; poor conditions give long slim hyphae — the organism reaching for better ground. Both are computable from the model.

Compared against *Geotrichum candidum*, *Streptomyces hygroscopicus* and *Penicillium chrysogenum*: l_{hgu} falls with D while the hyphal growth unit volume stays nearly constant.

7.7

7
Transport through the cell membrane

Passive diffusion sets the baseline

$$J = (D_{\text{mem}} / d_{\text{mem}}) K_{\text{par}} (c_a - c_b) = P (c_a - c_b)$$

Equation (7.64): the three factors collapse into the permeability coefficient P .

The membrane is 6–10 nm of bilayer, typically 50 wt% protein.

O_2 , CO_2 and undissociated acids cross — K_{par} 0.03 for HAc, 0.44 for HBu.

Hexoses have P four to five orders of magnitude lower.

H^+ , K^+ and Na^+ cannot cross by passive diffusion at all.

Example 7.10 · Facilitated transport

$$J = (D/d) \cdot c_t s_a / (K'_{eq}/K + s_a)$$

A complete analogue of Michaelis–Menten kinetics (6.8), with K'_{eq} the dissociation constant of the carrier–sugar complex.

The permease takes form E_1 outside and E_2 on the cytosol side.

Binding is at equilibrium at every point; all species diffuse at steady state.

Since $K \ll K'_{eq}$, the carrier route far outruns free diffusion.

It saturates, and other sugars inhibit it — mannose transport by glucose.

7.7.2 · Active transport

Symport

Lactose permease carries lactose together with a proton. The protons must then be pumped out at the expense of ATP, together with the lactate, to hold the internal pH near 7.

Uniport and antiport

Yeast hexose uptake is uniport, driven by free energy alone. In antiport the flow of one species is supported by the flow of another in the opposite direction.

Group translocation

The PTS system phosphorylates the sugar using PEP. High affinity — maximum rate even at very low s — and G6P cannot leak back out. Net yield still 2 ATP per glucose.

Example 7.11 · Organic acids uncouple

$$c_{b,\text{tot}} / c_{a,\text{tot}} = (1 + 10^{\text{pH}_b - \text{pK}_a}) / (1 + 10^{\text{pH}_a - \text{pK}_a})$$

Phenoxyacetic acid, $\text{pK}_a = 3.1$: P is $3.2 \cdot 10^{-6} \text{ m s}^{-1}$ undissociated against $2.6 \cdot 10^{-10}$ dissociated, so only the pH terms survive.

Undissociated acid diffuses in, dissociates, and the protons are pumped out.

At external pH 6.5 the acid accumulates 2.3-fold; at pH 5.0, about 100-fold.

The futile cycle costs $0.15 \text{ mmol ATP (g DW h)}^{-1}$ at pH 6.5 — about 7 at pH 5.0.

Lactic acid cultivations stop near 42 g L^{-1} , substrate still available.

Problems 7.1 – 7.5

Monod parameters from chemostat data · lactic acid inhibition · modelling the lac-operon · facilitated transport · phosphotransferase transport systems